

AB Power Meter Monitor — systemd Deployment Guide

Overview

The monitor runs in **headless mode** under systemd as a long-lived service (count=0, infinite loop). The `ab_meter_caller.py` wrapper is **not needed** for the service — `ab_power_meter_monitor.py` is invoked directly with `--headless --count 0`. The caller script is only useful when you want to drive the monitor from external Python code.

File Layout

<code>ab_power_meter_monitor.py</code>	← main script (run directly)
<code>ab_meter_caller.py</code>	← optional — only needed for external Python callers
<code>ab_stop.py</code>	← graceful stop helper
<code>abmeter-wrapper.sh</code>	← shell wrapper (flock + PID file + log tee)
<code>abmeter.service</code>	← systemd unit file
<code>abmeter_logrotate.conf</code>	← logrotate config (system mode only)
<code>install.sh</code>	← automated installer

Do I Need sudo?

Mode	Requires sudo?	Notes
System-wide (sudo <code>./install.sh</code>)	Yes	Creates <code>/etc/systemd/system/abmeter.service</code> , a <code>abmeter</code> system user, <code>/run/abmeter/</code> , and <code>/etc/logrotate.d/abmeter</code> . Service starts at boot for all users.
Per-user (<code>./install.sh --user</code>)	No	Installs to <code>~/.config/systemd/user/</code> . Service starts on login (or at boot if <code>linger</code> is enabled — prompted by installer).

Recommendation for a dedicated monitoring host: use system mode with `sudo`. For a developer workstation or shared host: use user mode.

Quick Start

System mode (sudo)

```
# 1. Put all files in one directory, then:
sudo ./install.sh

# 2. Check status
sudo systemctl status abmeter
sudo journalctl -u abmeter -f
```

User mode (no sudo)

```
./install.sh --user

# Check status
systemctl --user status abmeter
journalctl --user -u abmeter -f
```

How Single-Instance Prevention Works

Three independent layers ensure only one instance ever runs:

1. **systemd** `Type=simple` — `systemd` tracks the PID and will never launch a second `ExecStart` while the first is alive.
2. `flock --nonblock` in `abmeter_wrapper.sh` — the wrapper acquires an exclusive lock on `/run/abmeter/abmeter.lock` before doing anything. If the lock is already held the wrapper exits immediately with code 1.
3. **PID file** at `/run/abmeter/abmeter.pid` — written by the wrapper, removed on exit via a trap. Operators can read it with `cat /run/abmeter/abmeter.pid`.

Configuration

All polling parameters live in `abmeter_wrapper.sh`:

Variable	Default	Description
POLL_COUNT	0	Iterations (0 = infinite)
POLL_INTERVAL	30	Seconds between polls
RAM_PCT_LIMIT	70	Flush+clear TIME_SERIES_STORE at this RAM %
INSTALL_DIR	/opt/abmeter	Root of the installation

After changing the wrapper, reload the service:

```
sudo systemctl restart abmeter    # system mode
```

```
systemctl --user restart abmeter # user mode
```

IP lists and output formats are configured inside

`ab_power_meter_monitor.py` (the `IP_LIST`, `ENABLE_*` constants near the top of the file).

Graceful Stop

The service responds to two stop mechanisms:

```
# 1. systemd stop – sends SIGTERM; Python catches it and exits cleanly.
sudo systemctl stop abmeter

# 2. ab_stop.py – creates a STOP_COLLECTION sentinel file that the polling
#    loop checks every 0.5 s. The loop finishes its current poll cycle,
#    writes final outputs, then exits with code 0.
/opt/abmeter/venv/bin/python3 /opt/abmeter/ab_stop.py
```

Both paths write all enabled output files before exiting.

Restart Behaviour

```
Restart=on-failure    → auto-restart on crash or non-zero exit
RestartSec=15         → wait 15 s before restarting
StartLimitBurst=5     → give up after 5 failures in 2 minutes
StartLimitIntervalSec=120
```

A clean exit (code 0 from `ab_stop.py`) does **not** trigger a restart.

Log Access

```
# Live tail (system mode)
sudo journalctl -u abmeter -f

# Last 100 lines
sudo journalctl -u abmeter -n 100 --no-pager

# On-disk log files (also written by wrapper)
ls /opt/abmeter/ab_meter_output/logs/

# User mode
journalctl --user -u abmeter -f
```

Uninstall

```
# System mode
sudo systemctl stop abmeter
sudo systemctl disable abmeter
sudo rm /etc/systemd/system/abmeter.service
sudo rm /etc/logrotate.d/abmeter
sudo systemctl daemon-reload
sudo userdel abmeter          # optional: remove service account
sudo rm -rf /opt/abmeter      # optional: remove all data

# User mode
systemctl --user stop abmeter
systemctl --user disable abmeter
rm ~/.config/systemd/user/abmeter.service
systemctl --user daemon-reload
loginctl disable-linger "$(id -un)"  # if you enabled linger
rm -rf ~/.local/share/abmeter      # optional: remove all data
```

Troubleshooting

Symptom	Check
Service fails to start	<code>journalctl -u abmeter -n 50</code> — look for Python import errors or missing venv
flock: another instance running	<code>cat /run/abmeter/abmeter.pid</code> — stale PID file; delete it and restart
No output files written	Verify <code>ENABLE_CSV</code> , <code>ENABLE_FITS</code> etc. in <code>ab_power_meter_monitor.py</code>
HTTP poll failures	Check <code>IP_LIST</code> in the script; verify network reachability from the service user
Service restart loop	<code>StartLimitBurst hit; sudo systemctl reset-failed abmeter && sudo systemctl start abmeter</code>